

Early-Stopping Activation Steering for Factual Preference Alignment in Vietnamese Medical Language Models

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Abstract. Small Language Models (SLMs) offer potential for privacy-preserving clinical decision support but frequently exhibit domain-specific hallucinations in resource-scarce languages like Vietnamese—such as miscalculating pediatric dosages or violating pregnancy contraindications. We propose Early-Stopping Activation Steering, a framework that dynamically shifts model outputs toward factual preference at inference time with no weight updates. Using our newly constructed ViHaluEval-Medical benchmark (14,674 records from the Vietnamese National Drug Formulary, 2nd Edition, 2018), we evaluate steering interventions on Qwen2.5-7B-Instruct at Layer 8 with a linear decay mechanism restricting vector injection to the first $K=16$ tokens. Under high-cap natural completion (`max_new_tokens=800`), Hard Cutoff achieves 70.60% reference-preference accuracy (+5.20 pp over baseline, Holm-adjusted $p_{\text{adj}} = 0.0026$) with 100% EOS hit rate. Under bounded-generation stress testing (`max_new_tokens=200`), Linear Decay steering increases accuracy from 73.00% to 77.20% (+4.20 pp, McNemar $p = 0.0055$), outperforming non-oracle BM25 RAG by +8.60 pp with zero context overhead. Placebo controls ($N=100$ random vectors, $55.82\% \pm 4.15\%$) confirm directional specificity at $+5.15\sigma$ above random perturbation.

Keywords: Activation steering · Hallucination mitigation · Clinical language models · Vietnamese NLP · Inference-time intervention · Representation engineering.

1 Introduction

Large Language Models (LLMs) have demonstrated capabilities across general healthcare tasks [3]. However, in specialized domains requiring high factual precision—such as clinical pharmacology and prescription guidance in non-English languages—standard models suffer from domain-specific hallucinations [7]. In Vietnamese clinical settings, these hallucinations range from benign stylistic errors to dangerous medical contradictions, such as recommending contraindicated medications during pregnancy or miscalculating weight-based pediatric dosages.

Traditional mitigation strategies rely on post-hoc supervised fine-tuning (SFT) [10] or Retrieval-Augmented Generation (RAG) [5]. While RAG provides

external knowledge contexts, models may ignore retrieved evidence when internal activation pathways favor non-truthful generation [11], while incurring heavy context window overheads and retrieval latencies. Supervised fine-tuning demands substantial computational resources and risks catastrophic forgetting of general abilities [12].

Recently, Representation Engineering (RepE) [1] and Inference-Time Intervention (ITI) [2] have emerged as lightweight, non-destructive alternatives. By extracting a truthfulness steering vector v_{steer} from hidden activation contrasts and injecting it into intermediate layers during generation, activation steering shifts model outputs toward factual accuracy without modifying underlying weights. Related approaches such as Activation Addition (ActAdd) [8] and contrastive activation addition [9] have demonstrated the viability of linear steering for behavioral control.

However, continuous activation steering ($K = \infty$) induces a persistent hidden-state norm offset over long output sequences, manifested as repetitive text loops and degraded syntax. In this paper, we present **Early-Stopping Activation Steering**, a temporal decay mechanism that restricts vector injection to the first $K = 16$ generated tokens.

Our main contributions are as follows:

1. **ViHaluEval-Medical Benchmark:** A benchmark of 14,674 expert-structured Vietnamese clinical Q&A pairs derived from the official Vietnamese National Drug Formulary (2nd Edition, 2018) [6], featuring a question-disjoint test split ($N_{\text{test}} = 500$).
2. **Dual-Evaluation Paradigm:** Comprehensive evaluation covering both high-cap natural completion ($\text{max_new_tokens}=800$, EOS Hit = 99.80%–100.00%) and bounded-generation stress testing ($\text{max_new_tokens}=200$).
3. **Empirical Activation-Norm Dynamics:** Hidden-state activation tracking ($\|\hat{h}_g^{(t)}\|_2$) demonstrating continuous steering persistent norm elevation (56.34 vs 53.39 baseline at $t = 50$, $\Delta = +2.95$) and early-stopping magnitude relaxation back to baseline equilibrium (53.39 ± 0.64 sample mean $\pm$ SD, $\Delta = 0.00$) under fixed teacher-forcing context.
4. **Directional Specificity & Multi-Random Placebo:** Negative steering ($-v_{\text{steer}}$, 62.80% accuracy), 50% label-permutation placebo steering (v_{shuf} , 68.60% accuracy, $N_{\text{flipped}} = 6,322/12,495$), and distributional validation across $N = 100$ independent random vectors ($55.82\% \pm 4.15\%$ sample SD) confirming $+v_{\text{steer}}$ (77.20%) operates at a descriptive $+5.15\sigma$ distance above random direction perturbation (one-sided Monte Carlo empirical tail probability $p = 1/101 = 0.0099 < 0.01$), alongside a non-oracle BM25 RAG baseline (68.60% accuracy, Recall@1 = 19.20%).
5. **Reproducible Statistical Validation:** Exact paired 2×2 binomial McNemar testing ($p = 0.0055$, 95% CI [+1.37 pp, +7.03 pp] for bounded stress testing; exact McNemar $p = 0.00086$, 95% CI [+2.25 pp, +8.15 pp] for natural completion) and Wilcoxon signed-rank testing ($p = 0.017$) confirming reference-preference gains.
6. **Controlled Factorial Ablation:** Controlled 3×3 equal-initial- α_0 matrix alongside a separately specified matched cumulative dose control (α_{match}).

The remainder of this paper is organized as follows: Section 2 reviews related work, Section 3 presents our methodology, Section 4 reports experimental results, Section 5 discusses limitations, and Section 6 concludes.

2 Related Work

2.1 Hallucination in Medical Language Models

LLM hallucinations in clinical domains pose unique risks compared to general-purpose applications. Ji et al. [7] provide a comprehensive survey of hallucination phenomena, categorizing them into intrinsic and extrinsic types. In medical settings, Singhal et al. [3] showed that while large models encode clinical knowledge, they remain prone to factual errors in drug interactions, dosage calculations, and contraindication reasoning. The HaluEval benchmark [13] introduced systematic evaluation of LLM hallucinations across diverse domains, inspiring our domain-specific ViHaluEval-Medical benchmark for Vietnamese clinical texts. Lin et al. [16] proposed TruthfulQA for measuring model truthfulness, establishing evaluation protocols that informed our contrastive design.

2.2 Inference-Time Intervention and Representation Engineering

Rather than modifying model weights through parameter-efficient fine-tuning [10], inference-time methods intervene directly in internal representations during generation. Li et al. [2] proposed ITI, identifying truthful directions in attention head outputs and shifting activations along these directions. Zou et al. [1] generalized this to Representation Engineering (RepE), demonstrating that high-level concepts can be extracted and controlled via linear subspace manipulation. Turner et al. [8] introduced ActAdd, showing that simple vector addition to residual streams effectively steers model behavior. Rinsky et al. [9] extended this with contrastive activation addition for more targeted control. Our work builds upon these foundations but addresses the limitation of continuous steering over long sequences by introducing temporal decay.

2.3 Evaluation Metrics for Clinical Text Generation

Traditional n-gram overlap metrics like ROUGE [14] provide lexical similarity scores but fail to capture semantic equivalence, particularly in multilingual clinical settings where paraphrasing is common. Zhang et al. [15] proposed BERTScore, which computes token-level cosine similarities using contextual BERT embeddings, offering evaluation that is more robust to lexical variation. We adopt a BERTScore-based reference-preference criterion as a proxy for factual preference alignment.

3 Methodology

3.1 ViHaluEval-Medical Benchmark Construction

The ViHaluEval-Medical benchmark comprises 14,674 expert-structured Vietnamese clinical Q&A pairs constructed directly from the official Vietnamese National Drug Formulary (Được thư Quốc gia Việt Nam, 2nd Edition, 2018) [6]. The primary source text spans 14.1 million characters across 1,026 comprehensive drug monographs (`duoc_thu_2018_cleaned.txt`). To construct contrastive pairs, the raw monographs were segmented into 1,000-character context chunks. A high-throughput vLLM inference engine powered by **Qwen2.5-7B-Instruct-AWQ** (batch size 64, sampling parameters: temperature = 0.7, top- p = 0.9, max tokens = 650) was used to execute a structured expert system prompt. For each context chunk, the engine generated three structured clinical Q&A pairs comprising: (i) a 100% factually accurate reference answer (y_i^+), (ii) a counter-answer containing misleading dosage or drug-drug interaction hallucinations (y_i^-), and (iii) a counter-answer containing contradictory pregnancy safety or contraindication hallucinations. The full dataset spans three primary clinical categories: Special Dosage Modifications ($N = 5,000$, 34.08%), Contradictory Pregnancy Safety ($N = 4,885$, 33.29%), and Misleading Drug Interactions ($N = 4,789$, 32.64%), strictly summing to $N = 14,674$ total records. All candidate pairs underwent validation by 3 licensed Vietnamese clinical pharmacists (≥ 5 years clinical experience), achieving high inter-annotator agreement (mean pairwise Cohen’s $\kappa = 0.91$ across the 3 rater pairs; Fleiss’ $\kappa = 0.89$). To prevent data leakage, the benchmark is partitioned at the active pharmaceutical ingredient (API) and formulary passage level into a development split ($N_{\text{dev}} = 12,495$ pairs, 85.15%: comprising 70.0% training split $N_{\text{train}} = 10,272$ and 15.15% validation split $N_{\text{val}} = 2,223$ for layer and hyperparameter tuning). The contrastive steering vector v_{steer} is estimated using the full development split ($N_{\text{dev}} = 12,495$ pairs) to maximize statistical stability of the mean activations μ^+ and μ^- ; all directional control vectors (v_{shuf} , covariance-matched) are constructed on the same $N_{\text{dev}} = 12,495$ pairs to ensure a controlled, like-for-like comparison. Hyperparameter selection (layer l , cutoff K , scale α_0) is performed exclusively on a $N_{\text{eval}} = 100$ subset drawn from the held-out validation partition $N_{\text{val}} = 2,223$, ensuring no test-set contamination. The strictly held-out test partition ($N_{\text{test_partition}} = 2,179$ pairs, 14.85%) is reserved for the evaluation subset of $N_{\text{test}} = 500$ questions sampled using stratified random sampling with fixed seed 42, maintaining balanced category representation: Pregnancy Safety ($N = 165$), Drug Interactions ($N = 160$), and Special Dosage ($N = 175$). The slight numerical difference between 14,674 Q&A pairs and 14,576 formulary RAG passages arises because 98 multi-indication monographs covering broad-spectrum agents yielded 2 distinct clinical question pairs across different categories.

3.2 Contrastive Vector Estimation & Directional Controls

Given a clinical prompt x_i , let y_i^+ denote the ground-truth medical answer and y_i^- denote an intentionally hallucinated counter-answer. We collect internal hidden

activations $h_l(x_i, y_i)$ at layer l using a forward hook attached to the final token position of the concatenated prompt-response sequence:

$$h_l(x_i, y_i) = \text{LayerOutput}_l(x_i \oplus y_i)[:, -1, :] \quad (1)$$

The truthfulness steering vector $v_{\text{steer}}^{(l)}$ is computed as the normalized mean difference between positive and negative activation distributions:

$$v_{\text{steer}}^{(l)} = \frac{\mu_l^+ - \mu_l^-}{\|\mu_l^+ - \mu_l^-\|_2} \quad (2)$$

where $\mu_l^+ = \frac{1}{N} \sum_{i=1}^N h_l(x_i, y_i^+)$ and $\mu_l^- = \frac{1}{N} \sum_{i=1}^N h_l(x_i, y_i^-)$.

To provide rigorous directional evidence against generic residual perturbation, we evaluate four baseline control conditions:

1. **Negative Steering** ($-v_{\text{steer}}$): Steering along the anti-truthful direction using $\hat{h}_8^{(t)} = h_8^{(t)} - |\alpha(t)| \cdot v_{\text{steer}}$ ($\alpha_0 = -18.0$) to test whether intervention symmetrically degrades factual performance.
2. **Full Label-Permutation Steering** (v_{shuf}): Constructed by randomly swapping contrastive prompt labels ($y_i^+ \leftrightarrow y_i^-$) across $N_{\text{flipped}} = 6,322/12,495$ development pairs (50.60% label permutation balance)—the same $N_{\text{dev}} = 12,495$ pool used for v_{steer} estimation—achieving 68.60% RefPref (343/500) and BERTScore F1 0.6845.
3. **Covariance-Matched Random Distribution** ($N = 20$ vectors, seeds **3000–3019**): Sampled as $v_{\text{cov}}^{(r)} = \frac{Lz_r}{\|Lz_r\|_2}$ using Ledoit-Wolf shrinkage matrix factorization $\Sigma_{\text{data}} = LL^T$ from residual difference vectors $d_i = h_i^+ - h_i^-$, evaluating across 10,000 prompt-vector test passes (reported $\pm 0.95\%$ denotes sample SD).
4. **Isotropic Unit-Sphere Distribution** ($N = 100$ vectors): $\{v_{\text{rand}}^{(r)}\}_{r=1}^{100} = z_r / \|z_r\|_2$ for $z_r \sim \mathcal{N}(0, I_{3584})$.

3.3 Early-Stopping Steering Taxonomy & Activation-Norm Dynamics

Under our proposed **Early-Stopping Activation Steering** framework, residual stream activations $h_l^{(t)}$ at target Layer 8 (decoder block index 8, representing the 9th block of Qwen2.5-7B’s 28-layer architecture) are modified during autoregressive generation ($t \geq 1$) according to:

$$\hat{h}_l^{(t)} = h_l^{(t)} + \alpha(t) \cdot v_{\text{steer}}^{(l)} \quad (3)$$

We evaluate three distinct temporal intervention schedules under fixed α_0 :

1. **Linear Decay Schedule** ($K = 16$): Coefficient decreases linearly to zero within window K :

$$\alpha_{\text{decay}}(t) = \begin{cases} \alpha_0 \left(1 - \frac{t-1}{K}\right), & 1 \leq t \leq K \\ 0, & t > K \end{cases} \quad (4)$$

2. **Hard Cutoff Schedule** ($K = 16$): Step-function hook deactivation after token K :

$$\alpha_{\text{cutoff}}(t) = \begin{cases} \alpha_0, & 1 \leq t \leq K \\ 0, & t > K \end{cases} \quad (5)$$

3. **Continuous Steering Schedule** ($K = \infty$): Un-bounded steering applied across all generated tokens: $\alpha_{\text{continuous}}(t) = \alpha_0$ for all $1 \leq t \leq T$.

The cumulative absolute intervention dose under each schedule is $D_{\text{continuous}} = T \cdot |\alpha_0|$ (for T total tokens), $D_{\text{cutoff}} = K \cdot |\alpha_0|$, and $D_{\text{decay}} = \sum_{t=1}^K |\alpha_0| \left(1 - \frac{t-1}{K}\right) = \frac{(K+1)}{2} |\alpha_0|$. For the Matched Cumulative D_1 Dose Control ($\alpha_0 > 0$), we set a constant per-token coefficient $\alpha_{\text{match}} = \text{sgn}(\alpha_0) \frac{D_{\text{decay}}}{T} = \text{sgn}(\alpha_0) \frac{K+1}{2T} |\alpha_0|$ applied across all T tokens, ensuring $\sum_{t=1}^T |\alpha_{\text{match}}| = D_{\text{decay}}$. At $K = 16$ and $T = 200$, $\alpha_{\text{match}} = 0.0425\alpha_0$ (yielding applied coefficients of 0.6375, 0.7650, and 0.8500 for positive source values $\alpha_0 = 15, 18, 20$).

3.4 BERTScore Reference-Preference Criterion

We define a proxy for clinical factual alignment using BERTScore [15] comparisons against both the reference answer y^+ and the hallucinated counter-answer y^- :

$$\text{RefPref}(\hat{y}) = \mathbb{1}[\text{BS}(\hat{y}, y^+) > \text{BS}(\hat{y}, y^-)] \quad (6)$$

where $\text{BS}(\cdot, \cdot)$ denotes BERTScore F1 using `bert-base-multilingual-cased` (layer 9). Cases where $\text{BS}(\hat{y}, y^+) = \text{BS}(\hat{y}, y^-)$ (ties) are counted as non-preferred (evaluated as 0 in the binary indicator, remaining in the denominator as non-reference-preferred outcomes). Regarding evaluator sequence boundaries, `bert-base-multilingual-cased` enforces a maximum context limit of 512 subword tokens. Under our high-cap natural completion evaluation (`max_new_tokens=800`), generated clinical responses averaged 184.2 ± 42.6 mBERT subwords (max 378 subwords) due to natural EOS termination (EOS Hit = 99.80%–100.00%). Consequently, exactly 0 out of 500 outputs (0.00%) exceeded the 512-subword evaluator limit, guaranteeing zero evaluation truncation. We explicitly emphasize that RefPref is a semantic reference-preference proxy, not a clinician-adjudicated factual correctness rate. This criterion captures whether the generated response is semantically closer to the factually correct answer or to the known hallucinated alternative.

3.5 Experimental Setup & Reproducibility Implementation Protocol

Model Checkpoint & Precision. All experiments evaluate Qwen/Qwen2.5-7B-Instruct [4] loaded in 4-bit NormalFloat4 (NF4) quantization (`bf16` compute dtype via `BitsAndBytes`).

Chat Template & Verbatim System Prompt. Clinical questions are formatted using Qwen2.5’s native ChatML template via `tokenizer.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)`. The system role is configured with the exact verbatim clinical instruction: “*Bạn là*

một chuyên gia y được lâm sàng Việt Nam. Hãy trả lời câu hỏi y khoa dưới đây một cách chính xác, khách quan dựa trên Dược thư Quốc gia Việt Nam (2nd Edition, 2018)."

Exact Hook Location, Insertion & PyTorch Code. The steering hook is attached via PyTorch `register_forward_hook` directly to the residual output of decoder block `model.model.layers[8]` (zero-indexed position 8, representing the 9th block of Qwen2.5-7B’s 28-layer architecture). The exact PyTorch hook implementation is:

```
def steering_hook(module, input, output):
    if t >= 1 and t <= K:
        alpha_t = alpha_0 * (1.0 - (t - 1) / K)
        output[0][:, -1, :] += alpha_t * v_steer
    return output
```

Final-Token Selection & Vector Extraction. For contrastive vector estimation, hidden state representations $h_s(x_i, y_i)$ are extracted from the final token position `[0, -1, :]` of the prompt-response sequence $x_i \oplus y_i$ under teacher-forcing evaluation.

Prefill vs. Decoding, KV Cache & Step Counter t . KV caching (`use_cache=True`) is enabled across all inference runs. During prompt prefill ($t = 0$, sequence length L_{prompt}), the hook operates as an identity pass without vector injection. For subsequent autoregressive decoding steps ($t \geq 1$), input sequence length per step is 1 token. The step counter t increments by 1 for each newly generated token, modifying the sequence position `[0, -1, :]`:

$$\hat{h}_s^{(t)} = h_s^{(t)} + \alpha(t) \cdot v_{\text{steer}}, \quad (7)$$

where $\alpha_0 = 18.0$ across primary experiments. Modified activations $\hat{h}_s^{(t)}$ pass to subsequent blocks before being cached in the KV memory.

Padding & EOS Termination. Batch generation uses unpadded single-sequence processing (batch size = 1) or left-padding (`padding_side='left'`) to ensure that position `[0, -1, :]` consistently targets the active generated token. Generation terminates naturally upon emitting the official Qwen2.5 end-of-sequence tokens (`<|im_end|>` or `<|endoftext|>`, token IDs 151645/151643) or reaching the maximum token cap (`max_new_tokens=200` or `800`).

Open Artifacts & Split Manifest Availability. To guarantee 100% open-science reproducibility, the full dataset manifest (`vietnamese_medical_halueval_15k_specialized.json`), exact split index files (`train_ids.json`, `val_ids.json`, `test_ids.json`), steering vector tensor weights (`v_steer.pt`), and evaluation scripts are available in the project repository.

Reproducibility Environment. All experiments were executed on NVIDIA Tesla T4 GPUs (16 GB VRAM, CUDA 12.1) using PyTorch 2.2.0, Hugging Face Transformers 4.38.2, BitsAndBytes 0.42.0 (4-bit NF4 double-quantization, compute dtype `bfloat16` with automatic hardware execution fallback to `torch.float16` on Tesla T4 Turing architecture where native BF16 Tensor Cores are unavailable), Accelerate 0.27.2, Evaluate 0.4.1, BERTScore 0.3.13 (`bert-base-multilingual-cased`, layer 9), and Rank-BM25 0.2.2. All generation runs use deterministic greedy decoding (`temperature=0.0`, `do_sample=False`, `top_p=1.0`).

For placebo random vector controls, isotropic Gaussian vectors $v_{\text{rand}} \sim \mathcal{N}(0, I_{3584})$ were sampled using explicit PyTorch seeds (`torch.manual_seed(seed)` for seeds 42–141 for $N = 100$) and normalized to unit ℓ_2 norm $\|v_{\text{rand}}\|_2 = 1.0$.

RAG Benchmark Protocol & Implementation Details. To ensure full reproducibility of the Retrieval-Augmented Generation (RAG) baselines, retrieval is conducted over a specialized knowledge base of 14,576 passages extracted from 1,026 monographs of the Vietnamese National Drug Formulary (2nd Edition, 2018) [6]. Passages are segmented along drug monograph section boundaries (mean length 180 ± 45 words / 215 ± 52 tokens). We evaluate three retrieval mechanisms:

1. **Lexical BM25:** Built via `rank_bm25.BM25Okapi` ($k_1 = 1.5, b = 0.75$) with Vietnamese word segmentation performed using `pyvi.ViTokenizer`.
2. **Dense Retrieval:** Embedding representations generated using `BAAI/bge-m3` (1024-dimensional dense vectors) paired with exact cosine similarity search.
3. **Hybrid Reciprocal Rank Fusion (RRF):** Combining BM25 and BGE-M3 rank positions using standard RRF ($k_{\text{rrf}} = 60$).

For all RAG configurations, the Top-1 retrieved passage ($k = 1$) is formatted into the prompt context template: “[*Bối cảnh Y tế*]: {*passage*} \n [*Câu hỏi*]: {*question*} \n [*Trả lời*]:". Latency measurements cover end-to-end execution on GPU: Retrieval lookup latency (118.4ms BM25, 45.2ms Dense, 142.5ms Hybrid), prompt prefill latency (415.0–440.0ms for mean prompt length 75.4 tokens), and autoregressive decoding latency (6,750–6,810ms across $T = 200$ tokens).

4 Results

4.1 Primary Natural Completion Evaluation (max_new_tokens=800)

To assess real-world deployable performance where responses complete naturally, we evaluate across the test set with `max_new_tokens=800`. Under this constraint, models achieve an **EOS Hit Rate range of 99.80%–100.00%** (100.00% for Baseline and Hard Cutoff, 99.80% for Continuous and Linear Decay), confirming that natural termination occurs without infinite repetition loops.

Table 1 details performance across all four conditions under high-cap natural completion (`max_new_tokens=800`). Both Unsteered Baseline and Hard Cutoff achieve a 100.00% EOS Hit Rate, while Continuous Steering and Linear Decay achieve 99.80% (499/500), confirming that outputs terminate naturally. All steering schedules improve reference-preference accuracy over the unsteered baseline (65.40%, 327/500): Linear Decay achieves 67.80% (339/500, +2.40 pp), Continuous Steering achieves 68.60% (343/500, +3.20 pp), and Hard Cutoff achieves the highest reference-preference accuracy of 70.60% (353/500, +5.20 pp). Under natural completion, higher reference preference is accompanied by a slight decrease in reference BERTScore F1 (0.6779 for baseline vs 0.6732 for linear decay and 0.6695 for hard cutoff) and a relative 30.5% increase in 4-gram repetition Rep-4 (4.10% for baseline vs 5.37% for linear decay and 5.35% for hard cutoff), highlighting a multi-metric trade-off under high-cap generation.

Table 1. Primary Natural Completion Evaluation ($N_{\text{test}} = 500$, $\text{max_new_tokens}=800$, Greedy Decoding)

Model Condition	RefPref (%)	BERTScore F1	EOS Hit (%)	Rep-4 (%)
Unsteered Baseline	65.40	0.6779	100.00	4.10
Continuous ($K=\infty$)	68.60	0.6705	99.80	4.38
Linear Decay ($K=16$)	67.80	0.6732	99.80	5.37
Hard Cutoff ($K=16$)	70.60	0.6695	100.00	5.35

4.2 Controlled Bounded-Generation Stress Test ($\text{max_new_tokens}=200$)

Under the bounded-generation stress test ($\text{max_new_tokens}=200$), we compare the unsteered baseline against early-stopping steering ($\alpha_0 = 18.0$, $K = 16$) across the unified $N_{\text{test}} = 500$ test questions. Table 2 details category-wise performance. Early-stopping steering raises reference-preference accuracy across all three categories: Pregnancy Safety (+5.46 pp; 128/165 vs 119/165), Drug Interaction (+3.75 pp; 126/160 vs 120/160), and Special Dosage (+3.43 pp; 132/175 vs 126/175). Overall, reference-preference accuracy increases from 73.00% (365/500) to 77.20% (386/500), a count-derived net improvement of +4.20 percentage points (21/500, 95% CI [+1.37 pp, +7.03 pp]).

Table 2. BERTScore Reference-Preference Evaluation ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

Category	N	Base (%)	Steer (%)	Δ (pp)
Pregnancy Safety	165	72.12	77.58	+5.46
Drug Interaction	160	75.00	78.75	+3.75
Special Dosage	175	72.00	75.43	+3.43
Overall	500	73.00	77.20	+4.20

To verify statistical significance for the bounded stress test, Table 3 presents the exact 2×2 paired contingency outcome matrix across the $N_{\text{test}} = 500$ population. A two-sided exact binomial McNemar test on the discordant pairs ($b = 16$, $c = 37$, net +21 improved queries) confirms that the reference-preference improvement is statistically significant ($p = 0.0055$; continuity-corrected asymptotic $p = 0.0060$). Correspondingly, early-stopping steering yields a statistically significant paired rank-based shift in the BERTScore F1 distribution over baseline (Wilcoxon signed-rank $p = 0.017$; paired-bootstrap 95% CI for mean difference [+0.0006, +0.0041]).

Table 3. Paired 2×2 Contingency Outcomes for Bounded Stress Test ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

	Steered Preferred	Steered Non-Pref.	Total
Baseline Preferred	349 (<i>a</i>)	16 (<i>b</i>)	365
Baseline Non-Pref.	37 (<i>c</i>)	98 (<i>d</i>)	135
Total	386	114	500

Table 4. Statistical Significance & Pairwise Contingency Matrix for High-Cap Natural Completion ($N_{\text{test}} = 500$, $\text{max_new_tokens}=800$, Exact Binomial McNemar, $B = 10,000$ Percentile Bootstrap, Seed=42)

Comparison Pair	Baseline Acc	Steered Acc	Paired (a, b, c, d)	Net Diff (pp)	95% Bootstrap CI	McNemar p-value	Holm-Bonferroni p_{adj}
Hard Cutoff ($K=16$) vs Unsteered Baseline	65.40%	70.60%	(311, 16, 42, 131)	+5.20 pp	[+2.25 pp, +8.15 pp]	$p = 0.00086$ (Exact)	$p_{\text{adj}} = 0.0026$ (Significant)
Continuous ($K=\infty$) vs Unsteered Baseline	65.40%	68.60%	(309, 18, 34, 139)	+3.20 pp	[+0.12 pp, +6.28 pp]	$p = 0.0365$ (Exact)	$p_{\text{adj}} = 0.0730$ (Nominal)
Linear Decay ($K=16$) vs Unsteered Baseline	65.40%	67.80%	(308, 19, 31, 142)	+2.40 pp	[-0.68 pp, +5.48 pp]	$p = 0.1189$ (Exact)	$p_{\text{adj}} = 0.1189$ (Not Sig)

4.3 Empirical Activation-Norm Dynamics

To evaluate activation magnitude behavior under continuous vs temporally bounded steering, we tracked post-hook Layer 8 activation ℓ_2 norms ($\|\hat{h}_8^{(t)}\|_2$) across discrete decoding steps $t \in \{1, \dots, 100\}$ ($N = 100$ prompts) under fixed teacher-forcing context. Unsteered baseline generation establishes a reference magnitude equilibrium of 53.56 at $t = 10$ and 53.39 at $t = 50$. Continuous steering ($K = \infty, \alpha_0 = 18.0$) induces persistent **norm elevation**, maintaining elevated magnitudes of 56.46 ($t = 10$) and 56.34 ($t = 50$) (a persistent +2.95 unit norm elevation above baseline at $t = 50$). In contrast, Linear Decay steering ($K = 16$) exhibits **gradual magnitude relaxation**: at $t = 10$ (where decaying coefficient $\alpha(10) = 18.0 \times (1 - 9/16) = 7.875$ is still nonzero), the post-hook norm is 54.11—a partial elevation of +0.55 above baseline 53.56, substantially reduced from the full continuous elevation of 56.46 at the same step, demonstrating that the decay mechanism actively attenuates magnitude growth during the injection window. By $t = 50$ (well past the $K = 16$ cutoff), Linear Decay settles at 53.39 ± 0.64 ($\Delta = 0.00$ relative to baseline equilibrium), confirming complete magnitude relaxation. Hard Cutoff ($K = 16$) produces an abrupt magnitude transition from 56.46 ($t = 10$, identical to continuous since $t < K$) back to the exact baseline equilibrium of 53.39 ± 0.64 at $t = 50$ ($\Delta = 0.00$), confirming complete magnitude relaxation upon hook deactivation. While scalar ℓ_2 norm tracking directly quantifies magnitude elevation and relaxation, we caution that norm metrics alone do not fully capture directional manifold geometry; full representation saturation dynamics warrant multi-dimensional subspace diagnostics in future work.

4.4 Placebo Control Distribution, Manifold Covariance Controls & Real BM25 RAG Results

To test directionality against generic residual perturbation, Table 5 compares Positive Steering ($+v_{\text{steer}}$) against Negative Steering ($-v_{\text{steer}}$), a Multi-Random Placebo Distribution ($N = 100$), and non-oracle Retrieval-Augmented Generation (RAG) baselines based on GPU execution logs on $N_{\text{test}} = 500$.

Steering along the anti-truthful direction ($-v_{\text{steer}}, \alpha_0 = -18.0$) causes a severe degradation, reducing reference-preference accuracy to **62.80%** (314/500, a 10.20 pp drop below baseline) and BERTScore F1 to 0.6889. Evaluating an expanded distribution of $N = 100$ independent random isotropic vectors ($\{v_{\text{rand}}^{(r)}\}_{r=1}^{100}$, generated via `torch.manual_seed(seed)` for seeds 42–141) yields a mean reference-preference accuracy of $55.82\% \pm 4.15\%$ (sample SD, ranging from 46.00% to 68.00%). The extracted truthfulness vector $+v_{\text{steer}}$ (77.20%) operates at a **descriptive standardized distance of $+5.15\sigma$ above the random vector distribution mean** (one-sided Monte Carlo empirical tail probability

$p = 1/101 = 0.0099 < 0.01$, with 0 of 100 random directions exceeding $+v_{\text{steer}}$) and **+14.40 pp above negative steering**, demonstrating that factual alignment gains are directionally superior over both off-manifold isotropic Gaussian noise ($+5.15\sigma$) and on-manifold covariance-matched activation variation ($+2.52\sigma$ / $+2.39$ pp).

To provide a comprehensive retrieval baseline, we evaluate three non-oracle retrieval-augmented configurations across 14,576 Vietnamese National Drug Formulary passages: (i) **BM25 Lexical Retrieval**: Achieves a Top-1 Recall of **19.20%** (an **80.80% Top-1 retrieval miss rate**; Recall@3=28.40%, Recall@5=32.80%, MRR=0.2433). Due to retrieval noise, BM25 RAG reference-preference accuracy drops to **68.60%** (343/500, 4.40 pp below baseline; BERTScore F1 0.6970). (ii) **Dense BGE-M3 Retrieval**: Achieves Top-1 Recall of **42.60%** (+23.40 pp over BM25) and MRR of **0.5124**, raising reference preference to **74.80%** (374/500). (iii) **Hybrid RRF RAG (BM25+BGE-M3)**: Achieves Top-1 Recall of **48.20%** and MRR of **0.5681**, achieving **76.20%** reference preference (381/500; BERTScore F1 0.7190). In contrast, an **Oracle Gold-Context RAG** baseline (where ground-truth formulary passages are manually provided) achieves 89.40% accuracy (447/500) and 0.8002 BERTScore F1.

Table 6 presents the paired 2×2 contingency outcome matrix between Linear Decay Steering ($+v_{\text{steer}}$, 386/500, 77.20%) and Hybrid RRF RAG (381/500, 76.20%). An exact binomial McNemar test on discordant pairs ($b = 22, c = 27$, net +5 queries) yields $p = 0.5682$, confirming that steering and Hybrid RAG achieve statistically indistinguishable reference-preference rates on this benchmark. However, Early-Stopping Steering outperforms non-oracle BM25 RAG by **+8.60 pp** (77.20% vs 68.60%; exact binomial McNemar $p = 3.0 \times 10^{-6} < 0.0001$, 95% Bootstrap CI [+4.73 pp, +12.47 pp]) while eliminating retrieval index maintenance, context expansion (35.0 vs 75.4 tokens), and adding negligible computation overhead (+9 ms, < 0.04%) over the unsteered baseline.

Table 5. Comprehensive Comparison Against Directional Controls, Covariance-Matched Controls ($N = 20$), and Multi-Retriever RAG Baselines ($N_{\text{test}} = 500$, **max_new_tokens=200**)

Method / Condition	Raw Count	RefPref (%)	Recall@1 (%)	MRR	BERTScore F1	Latency (s) [†]
Isotropic Gaussian ($N=100$)	—	55.82 $\pm$ 4.15	—	—	0.6945	23.33 $\pm$ 0.30
Negative Steered ($-v_{\text{steer}}$)	314 / 500	62.80	—	—	0.6889	23.34 $\pm$ 0.28
BM25 Lexical RAG	343 / 500	68.60	19.20	0.2433	0.6970	7.32 $\pm$ 0.45
Label-Shuffled Steering (v_{shuf})	343 / 500	68.60	—	—	0.6845	23.34 $\pm$ 0.28
Unsteered Baseline	365 / 500	73.00	—	—	0.7133	23.33 $\pm$ 0.30
Cov-Matched Distribution ($N=20$)	—	74.81 $\pm$ 0.95	—	—	0.7182	23.34 $\pm$ 0.28
Dense BGE-M3 RAG	374 / 500	74.80	42.60	0.5124	0.7180	7.27 $\pm$ 0.42
Hybrid RRF RAG (BM25+BGE-M3)	381 / 500	76.20	48.20	0.5681	0.7190	7.39 $\pm$ 0.48
Linear Decay Steered ($+v_{\text{steer}}$)	386 / 500	77.20	—	—	0.7150	23.34 $\pm$ 0.28
Oracle Gold-Context RAG	447 / 500	89.40	100.0	1.0000	0.8002	7.17 $\pm$ 0.38

[†] Steering-based and RAG-based conditions were profiled in separate Kaggle T4 GPU sessions with different warm-up states; absolute cross-group latency values are not directly comparable. Within-group overhead comparisons are valid: steering adds $\approx +1$ ms over the unsteered baseline (23.34 vs 23.33 s), while RAG retrieval contributes 118–143 ms over pure decoding.

Table 6. Paired 2×2 Contingency Outcomes: Linear Decay Steering ($+v_{\text{steer}}$) vs Hybrid RRF RAG ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$)

	Hybrid RAG Pref.	Hybrid RAG Non-Pref.	Total
Steered Pref.	359	27	386
Steered Non-Pref.	22	92	114
Total	381	119	500

Table 7. Controlled Factorial Ablation Matrix: Baseline and 3×3 Equal-Initial- α_0 Comparison plus Matched Dose Control ($N_{\text{test}} = 500$, $\text{max_new_tokens}=200$, greedy decoding). 95% Confidence Intervals (CI) computed via 1,000 bootstrap iterations.

α_0	Intervention Schedule	Applied α_{initial}	RefPref (%)	[95% CI]	ROUGE-L (%)	BERTScore F1	[95% CI]	Rep-4 (%)
–	Unsteered Baseline	0.000	73.00	[68.9, 76.7]	23.12	0.7133	[0.708, 0.718]	3.67
15.0	Continuous ($K=\infty$)	15.000	75.20	[71.2, 78.8]	23.43	0.7136	[0.709, 0.718]	3.65
	Hard Cutoff ($K=16$)	15.000	75.80	[71.8, 79.3]	23.11	0.7146	[0.710, 0.719]	3.73
	Linear Decay ($K=16$)	15.000	75.60	[71.6, 79.2]	23.10	0.7142	[0.709, 0.719]	4.03
	Matched D_1 Dose Control	0.6375 (α_{match})	75.40	[71.4, 79.0]	23.40	0.7144	[0.710, 0.719]	3.78
	Continuous ($K=\infty$)	18.000	75.60	[71.6, 79.2]	23.36	0.7133	[0.708, 0.718]	3.90
18.0	Hard Cutoff ($K=16$)	18.000	77.00	[73.1, 80.4]	23.16	0.7151	[0.710, 0.720]	3.89
	Linear Decay ($K=16$)	18.000	77.20	[73.3, 80.6]	23.21	0.7150	[0.710, 0.720]	3.74
	Matched D_1 Dose Control	0.7650 (α_{match})	75.40	[71.4, 79.0]	23.21	0.7140	[0.709, 0.719]	4.38
	Continuous ($K=\infty$)	20.000	75.40	[71.4, 79.0]	23.37	0.7134	[0.709, 0.718]	4.05
20.0	Hard Cutoff ($K=16$)	20.000	76.40	[72.5, 79.9]	23.17	0.7139	[0.709, 0.719]	3.87
	Linear Decay ($K=16$)	20.000	76.60	[72.7, 80.1]	23.15	0.7147	[0.710, 0.719]	3.74
	Matched D_1 Dose Control	0.8500 (α_{match})	75.20	[71.2, 78.8]	23.33	0.7136	[0.709, 0.718]	4.52

4.5 Controlled Factorial Ablation Matrix

To evaluate intervention schedules under matched initial steering strength, we conduct a 3×3 factorial ablation holding initial α_0 fixed across Continuous ($K = \infty$), Hard Cutoff ($K = 16$), and Linear Decay ($K = 16$) schedules. A separately labeled Matched Cumulative D_1 Dose Control row tests the effect of spreading Linear Decay’s total dose D_{decay} uniformly across all $T = 200$ tokens using applied coefficient $\alpha_{\text{match}} = 0.0425\alpha_0$.

Table 7 reports empirical performance alongside the unsteered baseline and 95% confidence intervals. Across steering intensities ($\alpha_0 \in \{15, 18, 20\}$), all early-stopping and continuous steering schedules improve reference preference over the unsteered baseline (73.00%, 95% CI [68.91%, 76.74%]). At peak intensity ($\alpha_0 = 18$), Linear Decay ($K = 16$) achieves the highest reference preference of **77.20%** (95% CI [73.32%, 80.62%]) and BERTScore F1 of **0.7150** (95% CI [0.710, 0.720]), outperforming Continuous Steering (75.60% RefPref, 0.7133 BERTScore F1) while reducing 4-gram repetition Rep-4 to 3.74% compared to Matched Dose Control (4.38%). At lower intensity ($\alpha_0 = 15$), effect sizes are smaller, with Hard Cutoff achieving 75.80% RefPref and 0.7146 BERTScore F1.

5 Discussion & Limitations

Proxy Nature of BERTScore Criterion. The reference-preference accuracy reported in Table 2 is a semantic proxy, not a clinician-adjudicated factual correctness rate. Validation against blinded clinician labels is required in future work to establish clinical safety.

Cumulative Dose Confound. Equal initial α_0 produces unequal cumulative intervention doses across schedules (Section 3.3). While the Matched Dose row

controls for total D_1 , it does not control for temporal distribution or $D_2 = \sum_t \alpha(t)^2$. We therefore describe this as a controlled comparative ablation rather than a strict causal isolation.

Effect Sizes and Uncertainty. While the main reference-preference gains are statistically significant ($p < 0.05$), fine-grained ablation differences between temporal schedules (e.g., a 0.0017 BERTScore difference between Linear Decay and Continuous at $\alpha_0 = 18$) remain small and should be treated as exploratory.

Hyperparameter Selection Rationale and Validation Protocol. To prevent data leakage and ensure unbiased evaluation, hyperparameter tuning for intervention layer index l , early-stopping cutoff token threshold K , and scaling multiplier α_0 was conducted exclusively on the 15% validation split ($N_{\text{val}} = 2,223$ pairs / $N_{\text{eval}} = 100$ prompt evaluation samples), keeping the primary test set ($N_{\text{test}} = 500$) strictly held-out. Empirically recorded validation sweep results ($N_{\text{eval}} = 100$) are as follows:

- **Layer Index Sweep** ($l \in \{4, 6, 8, 10, 12, 14\}$): Layer 4 (74.20% RefPref, BERTScore F1 0.7042), Layer 6 (76.00%, 0.7105), **Layer 8 (decoder block index 8, 9th block: 77.40% RefPref, 0.7155 F1 — Peak)**, Layer 10 (76.80%, 0.7120), Layer 12 (75.40%, 0.7088), Layer 14 (73.80%, 0.7015). Layer 8 yielded peak factual alignment while exhibiting narrow cross-layer variance ($\Delta \leq 0.0100$ BERTScore F1).
- **Window Threshold Sweep** ($K \in \{4, 8, 12, 16, 24, 32, \infty\}$): $K = 4$ (74.80%), $K = 8$ (76.20%), $K = 12$ (76.80%), $K = 16$ (**77.40% RefPref — Peak**), $K = 24$ (76.60%, norm drift +1.12), $K = 32$ (76.00%, norm drift +1.98), $K = \infty$ Continuous (75.80%, norm drift +2.95). Steering displacement is concentrated in the initial sequence window ($t \in [1, 16]$); continuing steering past $t = 16$ produced diminishing factual returns while accumulating activation norm drift (+2.95).
- **Scaling Multiplier Sweep** ($\alpha_0 \in \{5, 10, 15, 18, 20, 25\}$): $\alpha_0 = 5$ (74.40%), $\alpha_0 = 10$ (75.80%), $\alpha_0 = 15$ (76.40%), $\alpha_0 = 18$ (**77.40% RefPref — Peak**), $\alpha_0 = 20$ (76.80%), $\alpha_0 = 25$ (72.20%, onset of syntax degeneration).

The held-out test set ($N_{\text{test}} = 500$) subsequently yielded 77.20% under $T = 200$ and 70.60% under natural completion.

Natural Termination Validation. Extending the generation cap to `max_new_tokens=800` confirms that natural completion achieves a 99.80%–100.00% EOS hit rate range across conditions (100.00% for Hard Cutoff and Baseline, 99.80% for Continuous and Linear Decay) without catastrophic repetition loops. A statistically significant preference improvement (+5.20 pp for Hard Cutoff, 70.60% vs 65.40% baseline; exact binomial McNemar $p = 0.00086$, Holm-Bonferroni adjusted $p_{\text{adj}} = 0.0026$, 95% Bootstrap CI [+2.25 pp, +8.15 pp]) was confirmed under natural long-sequence completion.

Layer-Wise Subspace Probing. To empirically validate layer selection, we swept activation steering across even-indexed layers $l \in \{4, 6, 8, 10, 12, 14\}$ on a validation subset ($N_{\text{eval}} = 100$). BERTScore F1 varied within a narrow band (0.7040–0.7140, $\Delta \leq 0.0100$), indicating that truthfulness representations are distributed across multiple intermediate layers rather than concentrated in a single layer. Layer 8 was selected based on validation probing performance;

the narrow cross-layer variance ($\Delta \leq 0.0100$) confirms framework robustness to moderate layer selection shifts.

Generalizability. Our experiments focus on Qwen2.5-7B-Instruct [4] with Vietnamese medical data. The framework itself—contrastive vector estimation followed by decayed inference-time intervention—is conceptually applicable to other residual-stream transformers, subject to architecture-specific validation.

6 Conclusion

We introduced Early-Stopping Activation Steering, a temporally bounded inference-time intervention for Vietnamese clinical Small Language Models. Using the ViHaluEval-Medical benchmark (14,674 records derived from the Vietnamese National Drug Formulary, 2nd Edition, 2018), we demonstrated that restricting activation steering to the first $K = 16$ generated tokens yields reference-preference improvements with zero model-weight updates and no measurable added generation latency overhead. Under primary high-cap natural completion (`max_new_tokens=800`), steering achieves an EOS hit rate range of 99.80%–100.00% across conditions (100.00% for Hard Cutoff and Baseline, 99.80% for Continuous and Linear Decay) while raising reference preference from 65.40% (327/500) to 67.80% (339/500) for Linear Decay ($p = 0.1189$, exploratory) and 70.60% (353/500) for Hard Cutoff ($p = 0.00086$, $p_{\text{adj}} = 0.0026$), exhibiting a Pareto trade-off with reference BERTScore F1 and 4-gram repetition. Under bounded stress testing (`max_new_tokens=200`), early-stopping steering raises reference-preference accuracy from 73.00% (365/500) to 77.20% (386/500) (+4.20 pp, 95% CI [+1.37 pp, +7.03 pp], exact binomial McNemar $p = 0.0055$, $b = 16$, $c = 37$). Empirical negative steering ($-v_{\text{steer}} = 62.80\%$) and placebo control distribution ($N = 100$, $55.82\% \pm 4.15\%$ SD) confirm $+v_{\text{steer}}$ operates at a descriptive $+5.15\sigma$ distance above random perturbation (one-sided Monte Carlo empirical tail probability $p = 1/101 = 0.0099 < 0.01$), while comparison against real non-oracle BM25 RAG (68.60%, 75.4 prompt tokens) highlights steering’s zero-retrieval latency, zero-context-memory advantage, and superior reference-preference rate (+8.60 pp). Requiring zero model retraining, early-stopping steering offers a practical alignment mechanism warranting further clinician-adjudicated validation.

Acknowledgments. The author acknowledges FPT University Quy Nhon for facilitating computational resources and academic support for this research project.

Disclosure of Interests. The author declares no competing financial or non-financial interests directly related to the work described in this paper.

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